



Aladdin's magic carpet: Navigation by in-air static hand gesture in autonomous vehicles

Xiaosong Qian , Wendy Ju & David Michael Sirkin

To cite this article: Xiaosong Qian , Wendy Ju & David Michael Sirkin (2020): Aladdin's magic carpet: Navigation by in-air static hand gesture in autonomous vehicles, International Journal of Human-Computer Interaction, DOI: [10.1080/10447318.2020.1801225](https://doi.org/10.1080/10447318.2020.1801225)

To link to this article: <https://doi.org/10.1080/10447318.2020.1801225>



Published online: 16 Aug 2020.



Submit your article to this journal [↗](#)



Article views: 27



View related articles [↗](#)



View Crossmark data [↗](#)



Aladdin's magic carpet: Navigation by in-air static hand gesture in autonomous vehicles

Xiaosong Qian ^a, Wendy Ju^b, and David Michael Sirkin^c

^aSchool of Design and Art, Beijing Institute of Technology, Beijing, China; ^bJacobs Technion-Cornell Institute, Cornell Tech, New York, New York, USA; ^cMechanical Engineering, Stanford University, Palo Alto, USA

ABSTRACT

This paper presents a novel and exploratory investigation of how users might control future autonomous vehicles with user-defined in-air static hand gestures. In the era of autonomous vehicles, how to support “driving” without steering control may become a key issue affecting user experience. As the navigation interface of future autonomous cars will be wholly dependent on GPS, and passengers will be unable to make manual control adjustments, verbal or gestural communication will become a primary means to assist them in vehicle control. We thus focus on gesture as an innovative solution for this “final 100 meters” problem in automated navigation. A study ($N = 24$) conducted in a full chassis simulator shows that hand gesture control is feasible for autonomous vehicle navigation. It further reveals that hand gestures are influenced by task types, local regional norms, and participant culture. In particular, the spatial region where gestures occur can affect execution time, gender can impact user preference and demand, and culture differences and priorities can affect adoption, ease of learning, and user comfort, influencing longer-term use.

1. Introduction

In the age of handheld devices, natural user interaction (NUI) using voice or gesture based interfaces has become familiar, and recent research shows that NUIs can improve user experience (Wigdor & Wixon, 2011). Unlike the mouse or keyboard, NUIs are better adapted to users' lifestyles (Norman, 2013). Hand gesture interaction, a common type of NUI, is commonly used in AR/VR applications, and has recently been integrated into the automobile domain by several vehicle manufacturers including BMW and Mercedes Benz to control non-driving-critical interfaces. Related research on hand gesture interaction (Lee et al., 2015; Ohn-Bar & Trivedi, 2014; Pfleging et al., 2012) has mostly focused on driver distraction. But with the rapid development of autonomous driving, driver distraction may no longer be a pressing issue for vehicles in the future.

Self-driving technology has largely focused on way-finding and road safety, and has largely overlooked passengers' basic need to reach these true destinations. Through the analysis of autopilot technology and user navigation behaviors, recent research has revealed several factors that are likely to cause the “final 100 meters” problem that affects user experience in navigation during autonomous travel:

- (1) Insufficient precision in client-side maps results in users' inability to accurately mark target locations. Since the navigation interface for future self-driving cars will be highly dependent on routes generated by GPS maps, inaccurate map data, unmarked route changes, and short-distance navigation could all be

sources of travel problems (Bonin-Font et al., 2008). For example, toward the end of taxi or shared rides, passengers often ask drivers to make minor route adjustments to reach their true destinations, which are different from those indicated on drivers' GPS maps.

- (2) Users' navigation needs are sometimes uncertain and dynamic (Kiefer et al., 2017). In manual driving, for instance, users often drive along streets to search for places of interest (e.g., restaurants or shops) or moving objects or people (e.g., waiting friends), and change their minds about their destinations. But the operational cost of entering or changing destinations in GPS maps is high (Terada et al., 2006), and it may be difficult to satisfy similar user needs with GPS maps that are preset for autonomous driving.
- (3) Users' cognition has an additional impact on navigation results. Map reading imposes a massive cognitive burden on users (Hegarty et al., 2006; Liben et al., 2002; MacEachren, 2015). The initial questionnaire for this investigation finds that users are used to making minor route adjustments to reach their true destinations, which are often slightly different from the positions indicated on GPS maps. For instance, GPS systems often announce that “you have reached your destination” while still in the middle of the road, and passengers' true destinations – building entrances or parking lots – can be located some distance away from the given street address. In addition, large

suburban communities and rapid sprawl in cities can challenge even seasoned Chinese drivers to accurately locate their destinations by GPS.

So how can we direct and navigate autonomous vehicles while facing such challenges? As autonomous cars will be highly dependent on GPS maps and routes, and passengers will be unable to make manual (steering) control adjustments, verbal or gestural communication become the primary means of assuring they reach their intended destinations. Voice interaction faces several challenges, including 1) recognition failures due to language or accent differences (Hou et al., 2010; Yan & Vaseghi, 2002), p. 2) confusion between commands and conversation (Ma et al., 2007), such as during a telephone call, and 3) intrusion from ambient noise (Kokubo et al., 2002). We thus focus on gesture, which avoids these drawbacks, and benefits from common behaviors that many cultures share (such as presenting a “palm up” to suggest a stop) (Holler & Stevens, 2007; Holler & Wilkin, 2009). Inspired by Aladdin’s Magic Carpet in Disney’s famous story, we found a new way for passengers to guide routes without use of the steering wheel (see Figure 1). Hand gesture is a simple and intuitive way of communication (Goldin-Meadow, 1999). Indeed, most people already have experience supporting driver navigation using hand gestures (Austin & Sweller, 2014, 2017). Our pre-questionnaire survey found that 83.3% of all participants had the experience of communicating with drivers by gestures. Thus, we launched an investigation into whether hand gestures could be applied to autonomous vehicles for navigation.

In order to design appropriate hand gestures that provide a better user experience in autonomous vehicles in the future, our research questions are:

- How do users perform in-air static hand gestures to navigate an autonomous vehicle?

- Which region in the vehicle will users prefer to performing hand gestures?
- Are there any differences in in-air static hand gesture caused by task familiarity, performance region, gender, or culture?

1.1. Related work

Hand gestures are commonly divided into static and dynamic types (Reifinger et al., 2007). Static gestures involve only hand reshaping, while dynamic gestures involve large-scale movement of the arm and/or hand. Depending on the recognition technology, static and dynamic gestures may have different impacts on recognition accuracy (Plouffe et al., 2015).

The related research on hand gesture interaction is generally oriented toward manual driving. Following the “eyes on the road, hands on the wheel” premise introduced in 2007 (González et al., 2007), how to reduce driving distraction during interaction remains a priority for researchers (Jahani et al., 2017; Pfleging et al., 2012; Pickering et al., 2007). One main research topic focuses on touch gesture. Burnett, et al. (Burnett et al., 2013) integrated swipe gesture interaction with an in-vehicle touch screen for control to reduce distraction; Chung, et al. (Chung & Rantanen, 2010) designed a steering wheel with two integrated touchpads to recognize gestures; and Reich, et al. (Reich et al., 2013) evaluated performance by comparing several touchless gesture interactions in the car’s cockpit. Another main research topic is to improve the recognition performance of hand gesture interaction. Rümelin et al. (Rümelin et al., 2013) designed a vision-based real-time hand gesture recognition system adapted to the complex environment in vehicles; and Yin, et al. (Yin & Davis, 2014) developed a real-time recognition system for continuous hand gesture interaction by tracking 3D hand posture.

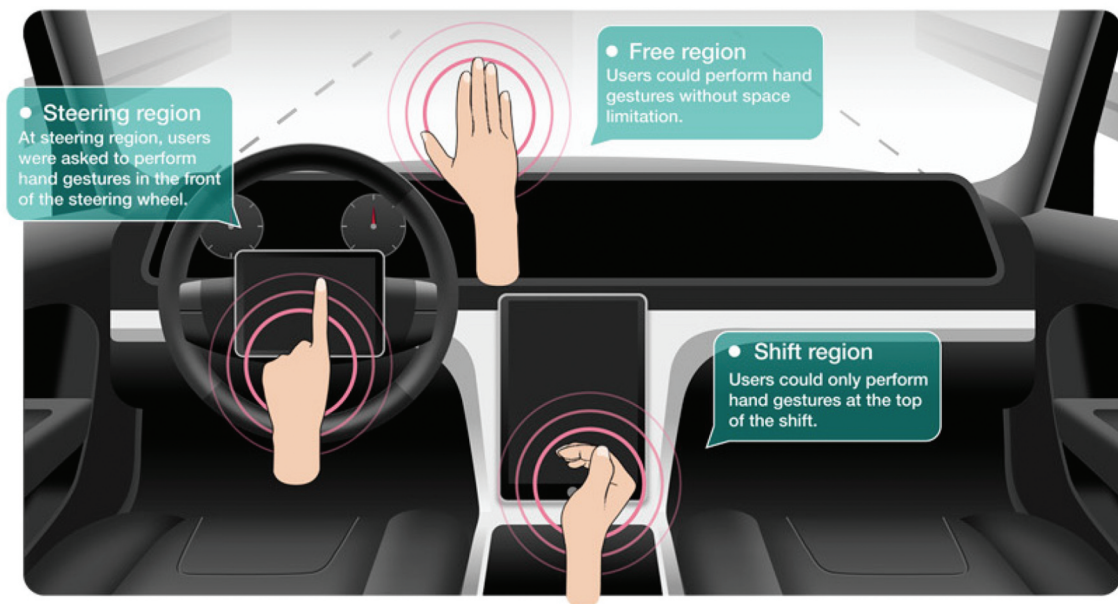


Figure 1. Inspired by Aladdin’s magic carpet, we found a new way for navigation without the steering wheel.

A proper taxonomy for hand gesture interaction in the automotive environment may help decrease cognitive load for both user and system (Scoditti, 2011; Stern et al., 2008; Wolf et al., 2011). Angelini (Angelini et al., 2014), et al. presented a taxonomy of user-elicited gestures performed on the surface of steering wheel; and Sun, et al. (Sun & Li, 2014) developed a design survey of in-air hand gestures for vehicle HMI.

In broader contexts, hand gesture interaction for navigation has been exploratively implemented in medical assistance devices for the disabled. Zeng et al. (Zeng et al., 2012) designed an intelligent wheelchair integrated with hand gesture interaction for navigation, and several studies (Chandarana et al., 2017; Fong et al., 2001; Mashood et al., 2015) show how hand gestures can be applied to object manipulation, such as drones or in video games.

For manually driven vehicles, the primary control operations are located around the steering and shifting regions, and prior research on gestural interaction confirms that most participants perform control gestures in these regions (Riener et al., 2013). Research also suggests that gender and culture can influence both hand gestures and driving behaviors (Archer, 1997; Dovidio et al., 1988; Harré et al., 1996; Kendon, 2004).

1.2. Method

1.2.1. Procedure

The experiment procedure consisted of a pre-questionnaire, an in-car simulator test, and a final questionnaire (see Figure 2).

In Part 1, participants were asked to independently complete a pre-questionnaire survey. The pre-questionnaire comprised several single-choice questions eliciting participants' navigation preferences while driving, including their opinions about autonomous vehicles, frequency of use of four different communication modes with drivers for navigation (paper map, speech, gesture, and digital map), and a question about situations where participants were willing to use gestures for navigation (such as not having a driver's license, parking, map failure, and short-distance adjustments to navigation).

In Part 2, the in-car test, participants were asked to perform in-air static hand gestures, using a single hand, to navigate the autonomous car through the simulated environment. The test was divided into three sequential tasks, based on the hand gesture performance region, for a between-groups design. Participants all began with the free region, where they could perform gestures without space limitation. The next two tasks were counterbalanced for order by performance region. Participants were evenly distributed by gender and culture for each of these orderings. The steering region was in front of the steering wheel, and the shift region was just above the shift lever. Participants could gesture only in the designated region during each of these two tasks. The shapes of their in-air static hand gestures were completely user-defined, without prior training or prompts.

Participants were required to observe road conditions by default. For each region, they were instructed to complete eight randomly ordered navigation tasks when the corresponding

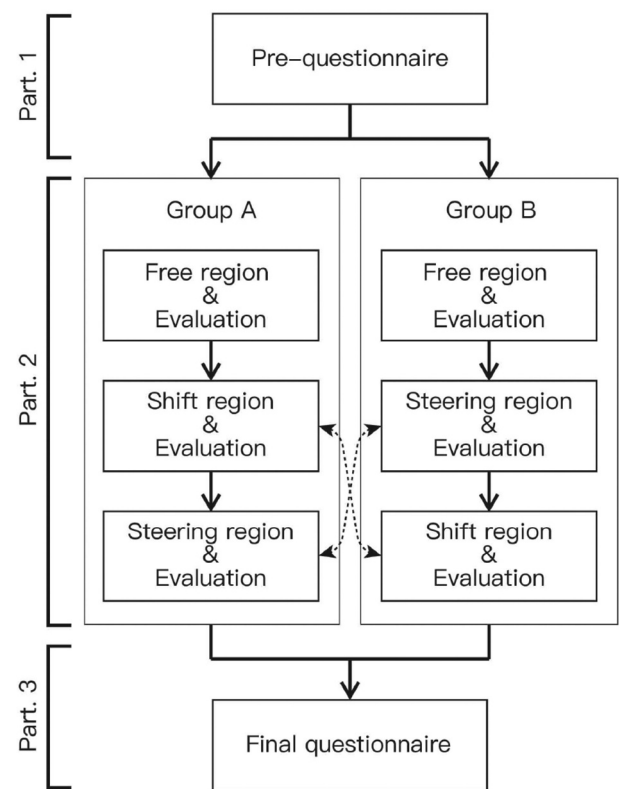


Figure 2. The experiment procedure consisted of a pre-questionnaire, an in-car test, and a final questionnaire. The shift and steering tasks of the in-car were counterbalanced for order by gesture region.

navigation task interfaces were displayed on the control panel with accompanying voice prompts. For each task, participants performed a static hand gesture within a specified task time (10 seconds) and maintained it until the end of the task time. The interval for each task was between 20 and 40 seconds. After completing eight randomly ordered navigation tasks in each region, participants took a short break and were asked to evaluate the difficulty of each task using a 5 point¹ (Dawes, 2008) Likert-type scale (extremely difficult, somewhat difficult, neither easy nor difficult, somewhat easy, and extremely easy).

In Part 3, participants were asked to complete a final questionnaire about their user experience preferences for the three performing regions, and their opinions on gesture navigation for autonomous driving (for example, whether they might like to “drive” using gestures, thought that two-handed gestures were needed, or preferred dynamic versus static gestures).

Following initial conversations with interaction design experts at Tier 1 automotive suppliers, we divided eight navigation tasks into three categories based on their frequency of occurrence in daily life (see Table 1).

1.2.2. Materials

The experiment took place in the driving simulator at the Automotive Innovation Facility at Stanford (see Figure 3). It is a whole chassis, full cab simulator, with original automobile components including instrument panel, steering wheel, brake and accelerator pedals, and adjustable seats with seat belt. The visual environment is displayed on a 12-foot high, 270-degree wraparound projection screen, with a separate screen for the

Table 1. Following the advice of Interaction Design (IxD) experts, eight tasks were divided into three categories by their frequency of occurrence in daily life.

Level ^a	Task
1	Go straight, turn left, turn right, stop
2	Slow down, back up
3	Turn around, pull over

^aLevel 1 = Most used, Level 2 = Commonly used, Level 3 = Rarely or not used.

rear view mirror. Four GoPro HERO cameras were set up inside the simulator to capture participants' behaviors during the test (see Figure 3). The research team remotely controlled the vehicle so that its driving appeared to be autonomous. All experiment materials and conversation were communicated in English for US participants and in Chinese for Chinese participants.

1.2.3. Participants

Twenty-four participants, including 12 US (6 males & 6 females) and 12 Chinese (6 males & 6 females), were recruited for the study. The average age was 35.3 ($SD = 14.3$) for US participants and 31 ($SD = 6.4$) for Chinese participants. Table 2 demonstrates the driving frequencies reported by both US and Chinese participants. Other than one Chinese participant who was unaware of his dominant hand, 11 Chinese participants and 12 US participants claimed to be right-handed. Additional requirements consisted of being in self-reported healthy condition, from 20 to 65 years old, being fully comfortable listening to, speaking, and reading English or Chinese, and having a full range of motion with the hands. Eleven Chinese and 12 US participants held driving licenses, although this was not a requirement, as the study was about experience using gesture, as a passenger, to inform a driver (or automated system) about navigation. All of the participants were confirmed as native US or Chinese citizens by a preliminary survey. Each participant was given a 25 USD Amazon gift card as compensation after the test.

t/w = time(s)/week, t/m = time(s)/month, NA = no driving experience.

1.3. Data processing

At the beginning of the experiment, participants were seated in the driver seat, and informed that the vehicle will drive autonomously through the simulated environment, controlled by a remote operator. We also asked that they disregard the vehicle's driving performance, so they would focus on the navigation task and their gestures.

Table 2. The driving frequencies for Chinese participants and US participants.

	Daily	4–5 t/w	2–3 t/w	1 t/w	2–3 t/m	1 t/m	<1 t/m	NA
CHN	3	2	2	1	0	0	3	1
US	5	1	3	1	1	1	0	0

Since we wanted to capture how cultural differences between China and the US might influence participants' actions, we asked them to use the same gestures during the in-car test that they would normally use when interacting with drivers in everyday situations.

A video coding team, comprising three experienced researchers, reviewed all of the videos to identify interactive hand gestures and execution time using Adobe After Effects CC (Millen, 2000). The video coding team defined “interactive hand gestures” as static gestures held by participants until the end of the task time, and defined “execution time” as the time between a prompt sound and when participants performed (completed) their interactive hand gestures (see Figure 4). The coding results of hand gestures ($N = 8$; $\alpha = 0.85$) and execution time ($N = 8$; $\alpha = 0.81$) for eight task types proved to be highly reliable. All data were analyzed using SPSS 19 (95% confidence interval).

2. Results

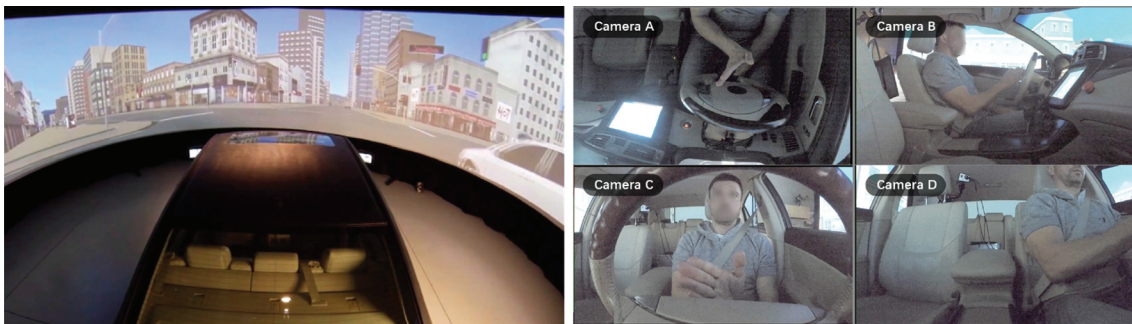
2.1. Interaction modalities

We have organized our observations of participant interactions around hand shape, and the consistency and inconsistency of hand gestures across individual participants and cultural groups.

2.1.1. Hand shape

Hand shape is a key variable affecting the semantics of hand gestures. Figure 5 demonstrates all of the hand shapes performed by US and Chinese participants during the test. In our study, hand gestures are classified into six categories based on the different fingers and hand forms: palm, fist, thumb, index finger, little finger, and roll. Although there are five types of palm form, there is no semantic difference by observation. The frequencies of hand shapes are shown in Table 3.

P = Palm, F = Fist, T = Thumb, I = Index finger, L = Little finger, R = Roll.

**Figure 3.** Four camera setup in an autonomous driving simulator. Camera A = Top view, Camera B = Side view, Cameras C & D = Front view.

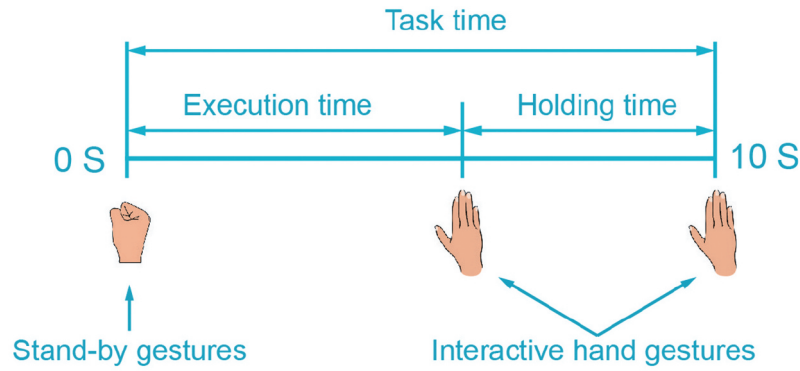


Figure 4. "Interactive hand gestures" and "execution time" defined by the video coding team.

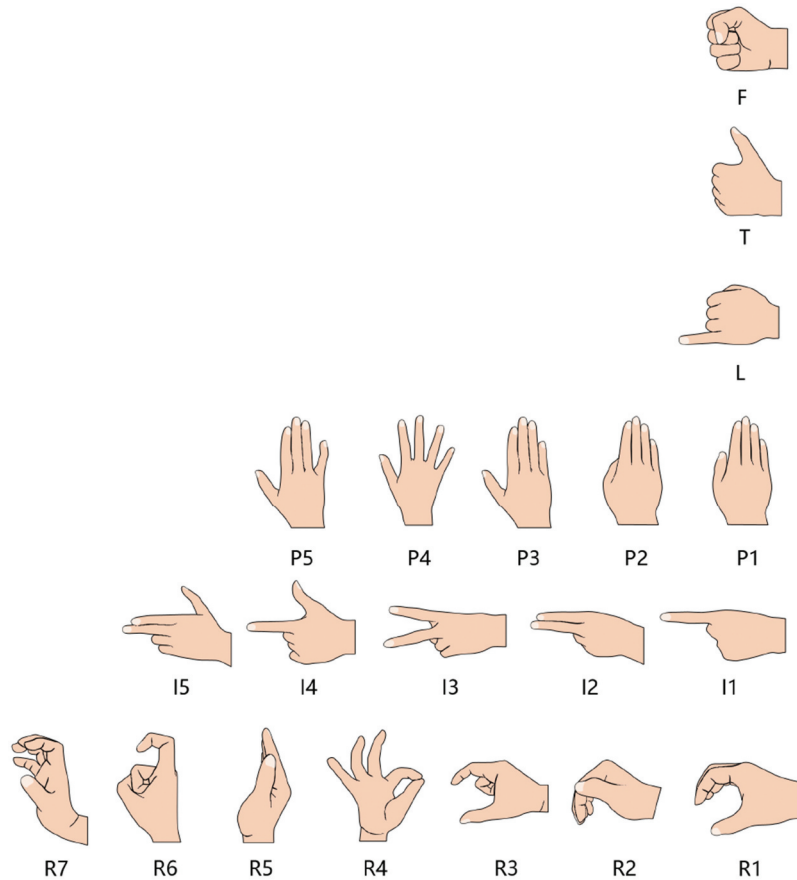


Figure 5. Six categories of user-defined hand shapes performed by participants in our research. P = Palm, F = Fist, T = Thumb, I = Index finger, L = Little finger, R = Roll.

2.1.2. Consistency

Preferred hand: The pre-questionnaire results show that all participants were right-handed, and, we observed that nearly all tasks were executed using the right hand. However, we also observed that gestures for the "turn right" and "pull over" tasks were often performed using the left hand, in both free and steering regions. Biomechanical limitations make it easier and more comfortable to bend one's hand in the direction of the palm (versus the back of the hand), specifically at the steering region. Thus, we find participants inclined to perform with their left hand those tasks regarded as having a right direction metaphor.

2.1.2.1. Hand shape. The results showed that both Chinese and American participants performed palm-shaped gestures most frequently (see Table 3), with 38.2% of gestures performed this way by US participants and 43.1% by Chinese. The other main gestures in the test feature pointing – including the thumb, index finger and little finger – where 37.1% were performed by US participants and 33.3% by Chinese. Furthermore, participants most frequently used 'I1' (in Figure 5) among the pointing-featured gestures. Hand shapes were basically not altered between the regions.

Table 3. Frequencies of hand shapes for each task (times).

	Go straight		Turn right		Turn left		Stop		Slow down		Back up		Turn around		Pull over	
	US	CHN	US	CHN	US	CHN	US	CHN	US	CHN	US	CHN	US	CHN	US	CHN
Free	P(5) I1(5) I2(1) I4(1)	P(8) I1(3) I4(1)	T(2) P(5) I1(4) I2(1)	T(2) P(6) I1(3) I4(1)	T(2) P(5) I1(4) I2(1)	T(1) P(6) I1(4) I4(1)	F(2) T(1) P(10) I1(1)	F(3) T(1) P(7) I1(1)	F(2) T(2) P(7) R7(1)	F(1) P(9) I1(1) R5(1)	T(5) P(1) R5(6) I1(1) R2(1) R5(3)	T(6) P(1) I1(1) R1(1) R2(3) R5(4) R6(1)	T(1) P(1) I1(1) R1(1) R2(5) R5(4)	T(1) P(1) I1(1) R2(5) R5(4)	F(1) T(2) P(7) I1(1) I3(1)	F(2) P(6) I1(2) R4(2)
Steering	P(5) I1(6) I2(1)	P(7) I1(4) I4(1)	P(3) I1(6) I2(1) I4(1) R5(1)	T(3) P(6) I1(2) I4(1)	P(3) I1(5) I2(1) I4(1) R5(2)	P(7) I1(3) I4(2)	F(3) P(9)	F(2) T(1) P(9)	T(1) P(8) R5(3)	F(1) P(8) I1(1) R5(2)	F(1) T(4) P(1) I1(1) R2(2) R2(3) R5(2)	T(5) P(1) I1(2) R1(1) R2(2) R5(2)	T(1) P(1) I1(1) R1(1) R2(5) R2(4) R5(3) R3(1) R5(2) R6(1)	T(2) P(1) I1(1) R2(5) R5(3)	F(1) T(2) P(5) I1(1) I3(1) I5(1) R5(1)	F(2) T(1) P(5) I1(2) R4(1) R5(1)
Shift	P(5) I1(5) I2(1) I4(1)	P(6) I1(6)	T(2) P(4) L(1) I1(3) I2(1) I4(1)	T(5) P(4) I1(2) I4(1)	T(3) P(1) I1(4) I4(1) R5(3)	T(2) P(1) I1(5) R5(4)	F(3) P(9)	F(3) T(1) P(8)	P(7) R3(1) R5(4)	F(1) P(9) R5(2)	F(1) T(4) I1(2) I4(1) R2(2) R5(2)	T(5) P(2) I1(1) R2(3) R5(1)	T(1) P(1) I1(1) I3(1) R2(7) R2(5) R5(1)	T(2) I1(1) I3(1) R2(7) R5(1)	T(2) P(7) I1(1) I3(1) I5(1)	F(2) P(6) I1(2) I2(1) R4(1)

2.1.2.2. Cultural influence. 1) Gun-type gesture: Some US and Chinese participants thought that gun-type gestures, such as ‘I4’ and ‘I5,’ might cause discomfort to others. 2) Stop: The vast majority of participants made a palm-forward, fingers-up gesture – the same one that police use. A few US and Chinese participants performed a fist-closed gesture, possibly inspired by the action performed by music conductors or military forces. Participants readily made the same gestures for the “stop” and “slow down” tasks, suggesting an ambiguity in their expression. 3) Turn around: Regardless of culture, many participants were not sure how to make a gesture for the “turn around” task. Unlike other orientation tasks, such as “go straight” or “turn left,” participants tried to demonstrate the path of the vehicle using a rolled hand, such as ‘R2’ and ‘R5,’ rather than indicate direction by gesture. 4) Pull over: “Pull over” was the other intractable task for most participants. Figure 6 displays the breadth of hand gestures performed by US and Chinese participants at the steering region. It shows that many participants, especially those from the US, performed right-pointing gestures, suggesting a “stop the car on the right” intuition when instructed to “pull over.”

2.1.2.3. Ergonomics. Ergonomics is an objective factor that can significantly impact hand gestures. There is a remarkable variation in feasibility considering the thumb, index finger, middle finger, ring finger and little finger. For instance, the index finger and thumb are more flexible than other fingers for the tasks of tapping or pointing. In our test, we found the index finger more frequently used than any other finger. The hand has more joints than any other part of the human body, which afford it great expressivity. However, joints also constrain the range of motion of the palm and fingers. During the test, numerous participants failed to make a palm gesture, and finally performed a rolled gesture such as “R5,” due to biomechanics (see Figure 7). This phenomenon occurred generally for the “back up” and “turn left” tasks.

2.1.3. Inconsistency

Task difficulty: For Level three tasks, we discovered that participants faced great challenges when performing gestures related to other tasks. Participants explained that they could only react by speculation: “I almost do not use such a gesture in my daily life.” This Lack of experience could be an indirect cause of their apparent inconsistency between “turn around” and “pull over.”

Cultural influence: 1) Pull over: US participants were inclined to perform right-oriented hand gestures. However, some Chinese participants performed the same gestures as “stop,” or a gesture that combined “turn right” and “stop” (see Figure 6). 2) Turn around: Some US participants demonstrated the path using rolled fingers, such as “R1,” “R3,” and “R6.” Conversely, no Chinese participants performed these gestures during the test. We note that ‘R6’ in particular already carries meaning for them – it represents the number 9 in Chinese – making its performance to indicate direction unlikely.

2.2. Execution time

We analyzed the significance of execution time using Kruskal-Wallis H test and the associated p -value (significance level: $p < .05$).

2.2.1. Region

Generally, execution time (which is reported in seconds) at the shift region ($M = 1.4$, $SD = 0.4$) was significantly less than at both the free region ($M = 1.9$, $SD = 0.5$) and steering region ($M = 1.8$, $SD = 0.4$) for each task, with $X^2(2, N = 576) = 141.6$, $p < .0001$ (see Figure 8.a).

According to Fitt’s law, the time to reach a target using a pointing device is related to the distance and the size of the target (MacKenzie, 1992). In this test, since the area of the free region (which had no size limitation) was much larger than the size of the steering region ($\approx 1,256 \text{ cm}^2$) and shift region ($\approx 1,000 \text{ cm}^2$), the displacement distance for hand gestures was a primary cause for the differences in execution time. Figure 9 shows that participants typically needed minimal



Figure 6. Hand gestures performed by US participants (left) and Chinese participants (right) at steering region for “pull over.”

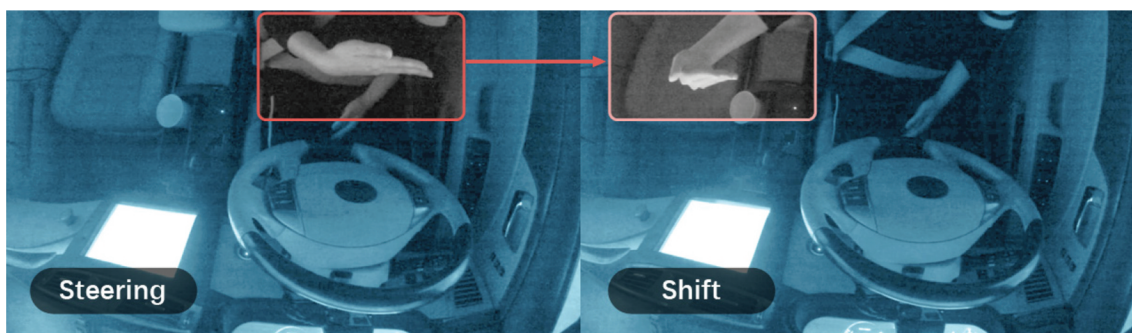


Figure 7. The hand gesture for the task of “turn left” at the steering and shift regions. The hand gesture changed due to ergonomics.

moving distance for gestures at the shift region relative to the other two. At the free region, participants normally raised their hands from a standby position to the height of their head or neck. In the steering region, they performed hand gestures at the height of the steering wheel, below the neck, in keeping with the test’s guidelines. At the shift region, starting at the standby position, participants could perform hand gestures over very small distances, even without any movement.

2.2.2. Task level

Execution time by participants on Level 1, Level 2 and Level 3 tasks was, respectively 1.5 ($SD = 0.4$), 1.8 ($SD = 0.4$) and 2.0 ($SD = 0.5$) (see Figure 8.b). Statistical analysis shows that participants generally had significantly different execution time on three levels of tasks, $X^2(2, N = 576) = 123.3$, $p < .0001$. Moreover, statistical analysis shows that there were significant differences in execution time of 3 levels regardless of free region ($X^2(2, N = 192) = 78.1$, $p < .0001$), steering region ($X^2(2, N = 192) = 38.1$, $p < .0001$) or shift region ($X^2(2, N = 192) = 55.3$, $p < .0001$).

We observed that participants could react using intuition or habit without thinking hard for Level 1 tasks. There was a strong consensus among participants and our IxD experts that hand gestures for Level 1 tasks are the most familiar in daily life. Hand gestures for Level 1 tasks are basically simple and easy, which supports less execution time. There are also gesture conventions for Level 2 tasks, although they are not as

common in daily life, so participants needed relatively more time to recall them. Level 3 tasks took the most time of all three categories, as participants spent more time speculating without experience.

2.2.3. Culture

Considering measures in the aggregate, we found no remarkable variation ($X^2(1, N = 576) = 2.0$, $p = .15$) between Chinese ($M = 1.7$, $SD = 0.5$) and US ($M = 1.7$, $SD = 0.5$) participants (see Figure 8.c). More specifically, we found that Chinese participants did have less execution time on deictic gestures (such as “go straight,” “turn right/left/around” or “back up”) than US participants. Conversely, US participants had less execution time on metaphoric/symbolic gestures such as “stop,” “slow down” or “pull over” (McNeill, 1992).

2.2.4. Gender

Consistent with the culture-based results, female participants ($M = 1.7$, $SD = 0.5$) were not significantly different from male participants ($M = 1.7$, $SD = 0.5$); $X^2(1, N = 576) = 1.1$, $p = .29$ (see Figure 8.d).

2.3. User evaluation

After each navigation section, participants completed questionnaires to evaluate how easy they found each task. As shown in Figure 10, these self-evaluation results are not completely consistent with the observed execution time results.

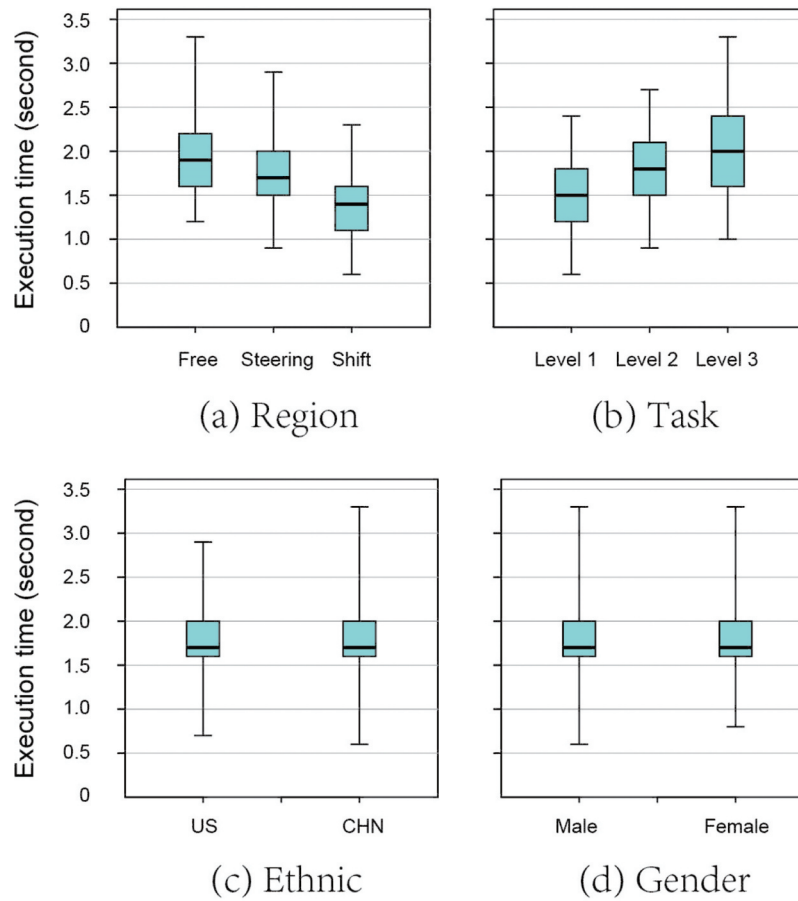


Figure 8. The statistical results of execution time by region, task, culture and gender.



Figure 9. The gesture movements at the free, steering, and shift regions. The movement at the shift region is more constrained than at the free and steering regions.

2.3.1. Region

In contrast to the significant difference found for execution time, there was no significant difference in perceived ease between gestures performed in free, steering, and shift regions; $X^2(8, N = 576) = 7.9, p = .44$ (see Figure 10.a).

Participants generally rated the free region as the easiest for performing hand gestures. Of 192 total tasks (8 (number of task types) * 24 (number of participants)), 131 were rated as somewhat easy or extremely easy at the free region. In contrast, 123 tasks at steering region and 120 tasks at shift region were rated as somewhat easy or extremely easy.

2.3.2. Task level

Participants considered most Level 1 tasks to be significantly easier than other tasks in the test, $X^2(8, N = 576) = 213.9, p < .0001$, while a majority considered Level 3 tasks to be the most difficult. There

was also a significant difference between evaluations of Level 2 and Level 3 tasks, $X^2(4, N = 288) = 12.1, p < .05$ (see Figure 10.b).

2.3.3. Culture

As shown in Figure 10.c, there was a significant difference between US and Chinese participants, $X^2(4, N = 576) = 12.8, p < .05$. Chinese participants rated the tasks for orientation (“go straight,” “turn right/left/around,” “back up”) easier than US participants. US participants were inclined to give higher evaluation to non-orientation tasks. We found that “turn around” was the most difficult task for US participants, while “pull over” was the most challenging task for Chinese participants.

2.3.4. Gender

Shown in Figure 10.d, there was a significant difference between male and female participants, $X^2(4, N = 576) = 20.4, p < .0001$.

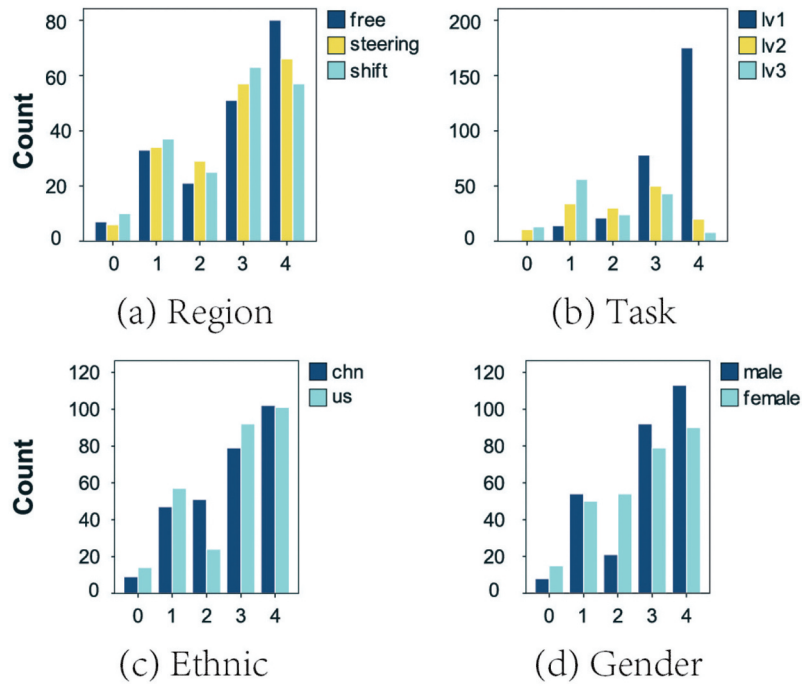


Figure 10. User evaluation by region, task, culture and gender. 0 = Extremely difficult, 1 = Somewhat difficult, 2 = Neither easy nor difficult, 3 = Somewhat easy, 4 = Extremely easy.

Chinese male participants rated task performance ease significantly different than Chinese female participants did, $X^2(4, N = 288) = 17.7, p < .005$. In particular, Chinese males rated over 44% of tasks as extremely easy, while Chinese females considered only 27% to be extremely easy. Meanwhile, Chinese female participants rated 25% of tasks as neither easy nor difficult, which was much higher than that of Chinese male participants. For US participants, there was a significant variation between males and females, $X^2(4, N = 288) = 15.3, p < .005$. The main difference was that US males rated over 40% of tasks as somewhat easy, while US females considered only 24% of tasks as somewhat easy.

2.4. Preference feedback

In addition to collecting data on performance and reports of ease of use, we asked participants about their preferences as potential consumers of gesture-based navigation systems.

2.4.1. Would you want a hand gesture navigation system?

Figure 11 shows Fisher's exact test finding of no notable difference by culture, $X^2(3, N = 24) = 3.4, p = .44$. Chinese participants generally showed more interest in and acceptance of the autopilot-related technology, compared with their US counterparts. For instance, over 83% of Chinese participants would like to have a hand gesture navigation system in their autonomous vehicles, while just half of US participants would.

Figure 11 also indicates that there was no significant difference between male and female participants, $X^2(3, N = 24) = 4.3, p = .24$. A hand gesture navigation system would seem to be more readily accepted by male participants, with over 83% of male and 50% of female participants showing interest.

2.4.2. Do you prefer dynamic or static gestures?

Figure 12 shows that both US and Chinese participants prefer to use dynamic rather than static gestures. In particular, US participants showed a greater preference for dynamic gestures than

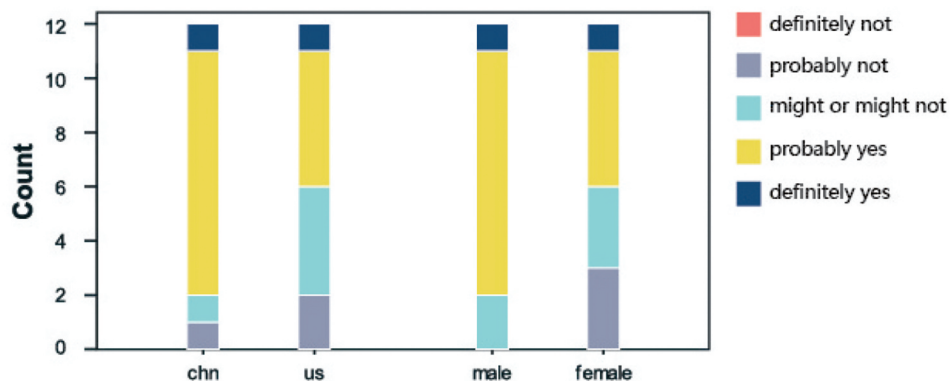


Figure 11. The results of how much one would want a hand gesture navigation system by participant group.

Chinese participants did (92% versus 67%). There was no statistically significant difference in culture, $X^2(3, N = 24) = 2.5, p = .56$.

The data further demonstrate that female participants were more likely to use dynamic gestures than male participants. Compared to only 8% of females, 25% of males preferred static to dynamic gestures. There was no statistically significant difference in gender, $X^2(3, N = 24) = 2.5, p = .56$.

2.4.3. Which was the most comfortable region?

The result of Fisher's exact test shows a dramatic difference among the three regions, $X^2(8, N = 72) = 31.6, p < .0001$. About 92% of participants had a positive report of comfort at the free region, which was much higher than at the steering (38%) or shift (42%) regions (see Figure 13).

There was a significant difference in comfort at the three regions, for both Chinese ($X^2(6, N = 36) = 17.4, p < .005$) and US ($X^2(8, N = 36) = 19.3, p < .01$) participants. US participants appeared to be more divided on the comfort of using the shift region than Chinese participants. While 50% of US participants gave a positive comfort

report for the shift region, 42% considered the experience to be negative. On the contrary, only 17% of Chinese participants expressed a dislike for their experience using the shift region. Half of Chinese participants rated comfort at the shift region neutral.

Additionally, Figure 13 shows that there was a tremendous difference between the three regions for both male ($X^2(6, N = 36) = 16.6, p < .005$) and female ($X^2(8, N = 36) = 15.7, p < .05$) participants. Males generally had higher comfort ratings at the three regions compared to females. But while males and females had highly similar comfort ratings for the free and steering regions, their ratings differed at the shift region, where 58% of males and 25% of females were keen on comfort, and 8% of males and 50% of females expressed aversion to their experience.

2.4.4. Which might influence your consumption more: Safety, price, or user experience?

Figure 14 shows a significant difference among safety, price, and user experience by Fisher's exact test; $X^2(6, N = 72) = 15.7, p < .01$. Clearly, safety is a primary factor affecting potential

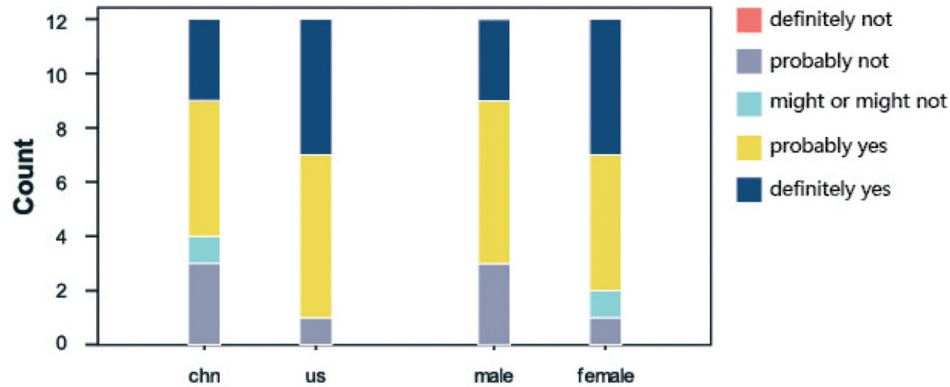


Figure 12. The results of user preference for dynamic or static gestures by participant group.

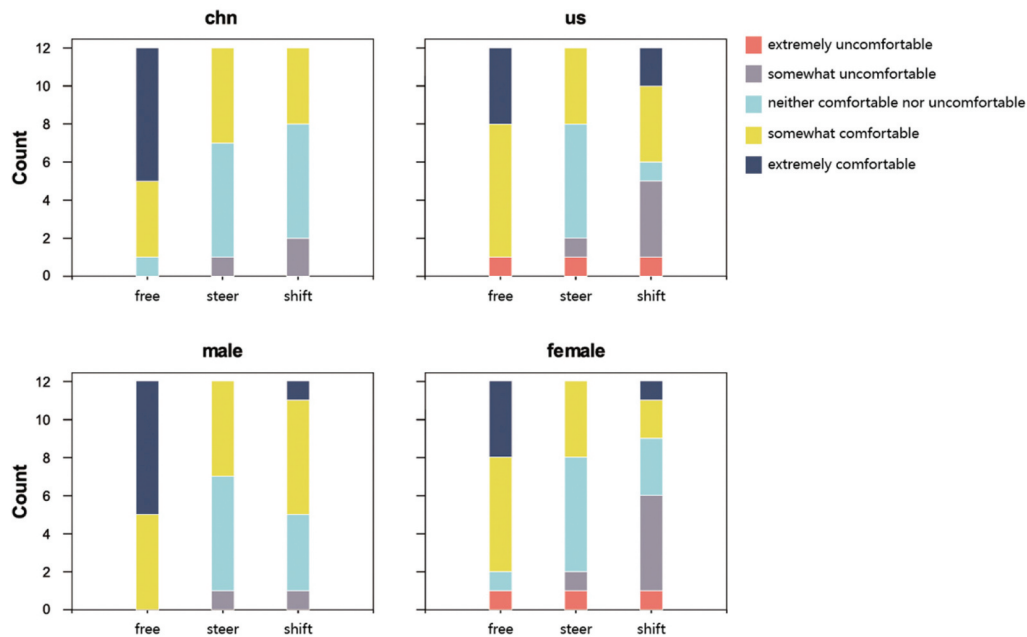


Figure 13. The results of comfort rating for 3 regions by participant group.

consumption, with 54% of participants considering safety as extremely important, versus 25% rating price and 21% rating user experience the same.

For Chinese participants, there was a dramatic difference among these three factors; $X^2(6, N = 36) = 12.6, p < .01$. Fifty percent rated safety as an extremely important factor, while only 16.7% believed that price would seriously affect their purchase. None regarded user experience as the most important factor.

For US participants, there was no significant variation between safety, price or user experience; $X^2(6, N = 36) = 8.6, p = .18$. Unlike the Chinese participants, 42% of US participants believed that user experience has a strong impact on their spending.

The study also reveals that Chinese participants were more sensitive to price. Over 83.3% of Chinese participants believed that price would affect their desire to purchase an autonomous vehicle with hand gesture navigation system, while only 41.7% of US participants regarded price as a key factor.

There was no dramatic variation among the factors for male participants; $X^2(6, N = 36) = 8.1, p = .21$, but there was a significant difference for female participants; $X^2(4, N = 36) = 9.2, p < .05$. One-third of males valued user experience as an unimportant factor influencing their aspirations, while 58% admitted that their purchase would be strongly affected by the user experience. All females regarded user experience as a key factor in their purchase.

2.4.5. Which is the most important feature of the system: Accuracy, speed, or comfort?

A Pearson chi-square test shows that there was a significant difference among recognition accuracy, recognition speed and

user comfort; $X^2(4, N = 72) = 23.8, p < .0001$. Additionally, regardless of culture or gender, the high-to-low order of importance was always recognition accuracy, recognition speed and user comfort (see Figure 15).

Overall, 67% of US and Chinese participants, and 75% of male and female participants, ranked recognition accuracy as a top priority. They ranked recognition speed as having secondary importance, slightly more than user comfort.

2.4.6. What other suggestions can you offer about the system?

Participants shared some additional opinions on the gesture system at the end of the test (see Table 4). Their reported needs suggest that safety, and the convenience of interaction, were their main concerns.

Comments included “It’s great that the system can learn or adapt to my own gestures, but for safety, I think if it had some standard gestures, that should be helpful. It can detect number and direction of fingers and distinguish my hand from that of other people in the car.” Also, “The hand gesture navigation system would be comprised of a set of standard hand signals to eliminate ambiguity. It would be nice to have voice command in addition to hand gestures.”

3. Discussion

We present several insights and design guidelines, based on our test data and observations.

3.1. Hand gestures could be an important solution for navigation

The pretest questionnaire revealed that participants were inclined to support driver navigation using software such as

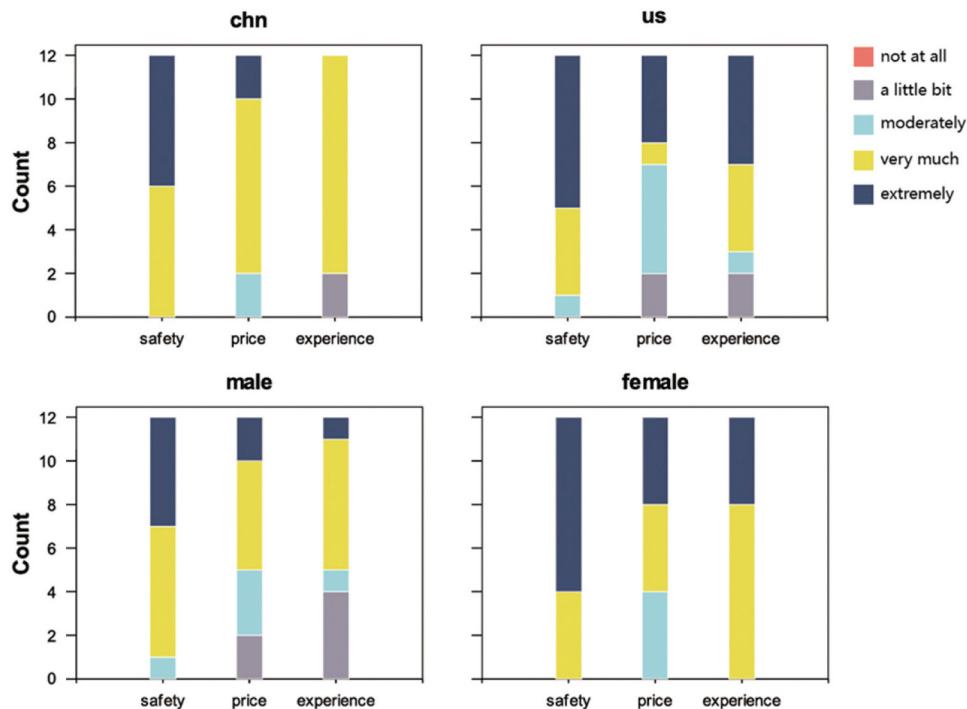


Figure 14. The results of the factors affecting consuming desires for gesture navigation systems by participant group.

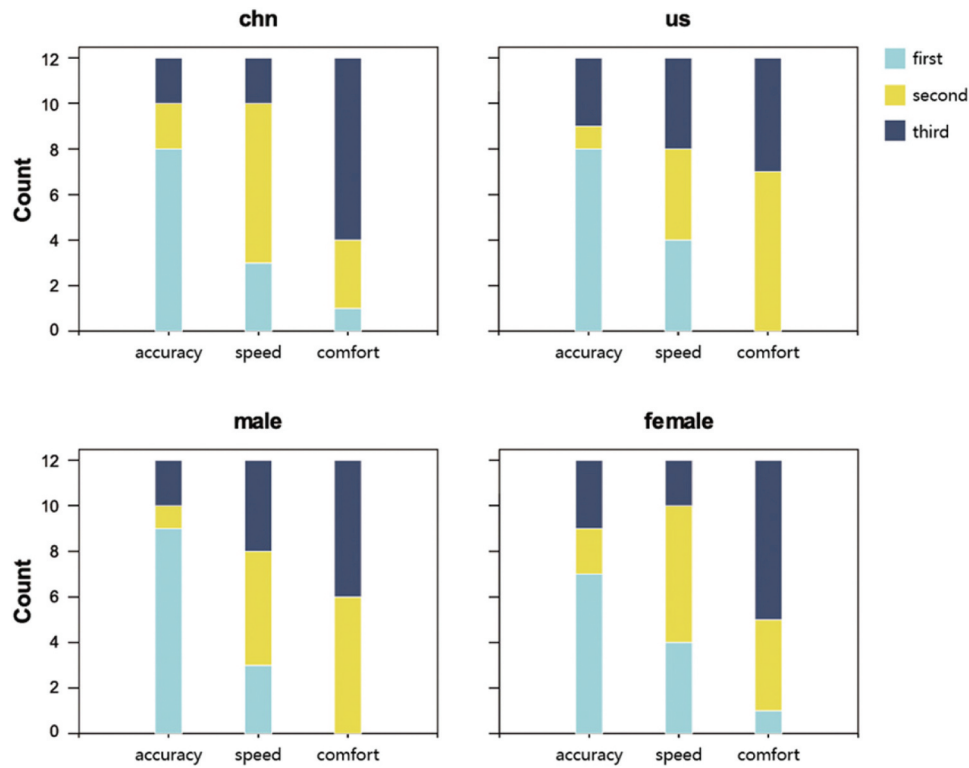


Figure 15. The results of the high-low sorting of recognition accuracy, recognition speed, and user comfort by participant group.

Table 4. Participants' additional needs for a gesture navigation system.

User needs	Frequency (N)
The addition of voice assistance	7
A standard set of hand gestures	5
Unambiguous interactive feedback	2
User identification	2
Child lock-out	1

Google Maps. However, common inconveniences such as insufficient map data accuracy or network failure can pose severe challenges to this form of wayfinding. For example, choosing a preferred parking lot, or a particular spot, is a challenge that drivers frequently face, and it is influenced by changing traffic conditions around the target destination. When picking someone up, an ideal parking spot would be as close as possible to that person's location. However, since drivers cannot obtain detailed traffic information through the GPS map beforehand, they have to delay such decisions until they approach the target destination. Therefore, speech and gesture will remain important and irreplaceable means of navigation for most people when interacting with GPS routing systems (Pfleging et al., 2012). Questionnaire results also found that both US and Chinese participants preferred to navigate using hand gestures when software failed, or over short distances.

3.2. Static gestures might be better than dynamic gestures for navigation

While participants preferred dynamic gestures, our study and prior research suggest that static gestures could be better suited to the autonomous driving context:

- Static gestures have less performance time than dynamic gestures. Dynamic gestures are composed by a series of static gestures (Kendon, 2004), which take more time to perform in sequence than static gestures alone.
- Static gestures cause less distraction. We observed that dynamic gestures block more of participants' visual field, and more easily distract their attention while searching for places of interest or moving objects (Bragdon et al., 2011; Pickering et al., 2007).
- Static gestures could be easier to standardize (Bau & Mackay, 2008).
- Static gestures could be easier to learn (Bau & Mackay, 2008; Bhuiyan & Picking, 2011).
- Static gestures could be easier for continuous movement or action. For example, users could readily flip their palm upward to change the command from 'go straight' to 'stop.'
- Static gestures require less performance space than dynamic gestures (Pickering et al., 2007). This will help reduce the area of gesture collection, which can reduce users' anxiety about privacy and disclosure (Demiris et al., 2009).
- Current research on gesture recognition shows that static gestures can be recognized faster and more accurately than dynamic gestures (Deo et al., 2016; Li et al., 2017; Singha et al., 2018; Yin & Davis, 2014; Zhuang et al., 2017). The result of our pre-questionnaire survey also reveals that participants considered recognition accuracy and recognition speed to be more important than (their own) user comfort in an autonomous driving context.
- Less execution time and faster recognition will provide immediate, timely interactive feedback which could

effectively help reduce users' anxiety in the process of interaction (Feng & Donmez, 2013).

It should be noted that gesture interaction based on static gestures may not be the final design choice, as adoption will depend on how well the system both functions and supports users' preferences. As the recognition speed and accuracy of dynamic gestures continue to improve, dynamic gestures may become a more feasible alternative.

3.3. The shift region could be better than other regions for hand gestures

Participants ranked their experience performing gestures at the shift region lower than at other regions. Still, we find several advantages to recommend this region for hand gesture navigation:

- Our analysis shows that the execution time for each task at the shift region is significantly less than at other regions.
- Performing gestures at the shift region blocks less of participants' visual field and causes less distraction to their attention.
- The user experience at the shift region could be improved by using carefully-designed hand gestures (S. S. Wu et al., 2016).
- Some hand forms, such as gun-type or thumb-down gestures, might cause discomfort to pedestrians or other passers-by through the front windshield (Azevedo et al., 2005). We observed that gestures performed at the shift region were effectively invisible from the outside of the vehicle.

Forster's overview of four years of research on interfaces for partially automated vehicles (Forster et al., 2018) reports that users with good interactive performance on a tested system frequently gave low preference scores to the system (preference-performance dissociation). In our study, we observed a similar effect. While participants' interaction performance (execution time) at the shift region was much better than at the free region, their evaluation scores for the shift region were comparatively lower. To resolve this issue, developing and testing well-designed gestures, and matching a potentially broader set of performance and self-report measures, will be the main focus of our future research.

3.4. Culture differences should be considered in gesture design

Across the study, results show that behaviors for navigation – especially those that form and project hand shapes – are influenced by cultural differences. Improper gesture design could not only cause discomfort to users and observers but also increase learning cost. As noted earlier, Chinese users would be most confused if they were instructed to perform the 'R6' gesture (see Figure 5) to turn the vehicle around. And such cultural distinctions in hand gestures are common worldwide. Follow-up research on user-defined hand gestures can make a great contribution to standard gesture design by

eliminating the cultural conflicts that can occur between regions or nations.

3.5. Safety must be a serious consideration

Our questionnaire results showed that all participants expressed great concern over the safety of using gesture navigation systems. Since there is no physical interaction interface, gesture operations could be easily interrupted by other actions – such as expressive conversation – or other in-car users. And unlike gesture control for multimedia systems, incorrect operations such as hand gesture recognition errors, child misuse, and untimely operational feedback can have huge negative impacts on (immediate and surrounding) vehicle operators and occupants (Jæger et al., 2008). Although fully autonomous driving could greatly improve driving safety overall, safety should always be the first priority of a gesture navigation system. We look forward to follow-up research focusing on approaches that ensure system safety.

4. Conclusion

In this paper, we explored a novel navigation interaction based on in-air hand gestures for autonomous vehicles. Without steering control or digital maps, autonomous vehicles could be "driven," like Aladdin's magic carpet, using hand gestures. By exploring user-defined gestures, the study helps us understand how people might use such gestures to navigate future vehicles. It also shows how hand gestures are influenced by the navigation task, performance region, and user culture. To improve the user experience, we suggest that designers consider cultural differences when developing control gestures. The study also reveals that gender has no significant impact on performance, but it may influence user preference and demand.

Although our study provides an interesting and feasible proposal for navigation applied to autonomous vehicles, it also has limitations. First, our user sample cannot fully represent the diversity of possible user personas. Further differences in hand gesture use due to experience, personality, sensory or mobility impairment, will need to be detected through in-depth studies. Second, user-defined hand gestures are not the best solution given current technology (H. H. Wu et al., 2019), but are rather a goal for the future. We found that some user preferences, such as the gesture interaction areas, may conflict with their interactive performance. Studies (Li et al., 2017; Singha et al., 2018; Yin & Davis, 2014; Zhuang et al., 2017) have found that existing systems do not perfectly recognize hand gestures, so given the first priority of safety, it is important to design new gestures that can be quickly and accurately recognized by near-future systems.

The broader gesture control approach has limitations as well. During manual driving, drivers can sense road feedback through the steering wheel and pedals, which helps them moderate their actions. Without tactile feedback, drivers using hand gestures may over-control the system or not recognize performance mistakes. Our questionnaire results and recent work (Jaschinski et al., 2016) suggest that we

include visual and audible feedback during gesture control, which may provide another direction for future work.

We hope that our findings will support the design of a new series of standard hand gestures for autonomous vehicles that can be easily learned and mastered by non-technical users.

Note

1. A preliminary test found that some participants had difficulty distinguishing the text descriptions for a 7 point scale.

Acknowledgement

Special thanks to Rebecca Currano, David Miller, Mishel Johns and Larry Cathey for their help with the simulator setup and data analysis.

ORCID

Xiaosong Qian  <http://orcid.org/0000-0001-9569-9528>

References

- Angelini, L., Carrino, F., Carrino, S., Caon, M., Khaled, O. A., Baumgartner, J., Sonderegger, A., Lalanne, D., & Mugellini, E. (2014). Gesturing on the steering wheel: A user-elicited taxonomy. In *Proceedings of the 6th International Conference on Automotive User Interfaces and Interactive Vehicular Applications* (pp. 1–8). New York, NY: ACM.
- Archer, D. (1997). Unspoken diversity: Cultural differences in gestures. *Qualitative Sociology*, 20(1), 79–105. <https://doi.org/10.1023/A:1024716331692>
- Austin, E. E., & Sweller, N. (2014, June). Presentation and production: The role of gesture in spatial communication. *Journal of Experimental Child Psychology*, 122, 92–103. <https://doi.org/10.1016/j.jecp.2013.12.008>
- Austin, E. E., & Sweller, N. (2017). Getting to the elephants: Gesture and preschoolers' comprehension of route direction information. *Journal of Experimental Child Psychology*, 163, 1–14. <https://doi.org/10.1016/j.jecp.2017.05.016>
- Azevedo, T. M., Volchan, E., Imbiriba, L. A., Rodrigues, E. C., Oliveira, J. M., Oliveira, L. F., Lutterbach, L. G., & Vargas, C. D. (2005). A freezing-like posture to pictures of mutilation. *Psychophysiology*, 42(3), 255–260. <https://doi.org/10.1111/j.1469-8986.2005.00287.x>
- Bau, O., & Mackay, W. E. (2008). OctoPocus: A dynamic guide for learning gesture-based command sets. In *Proceedings of the 21st annual ACM Symposium on User Interface Software and Technology* (pp. 37–46). New York, NY: ACM.
- Bhuiyan, M., & Picking, R. (2011). A gesture controlled user interface for inclusive design and evaluative study of its usability. *Journal of Software Engineering and Applications*, 4(9), 513. <https://doi.org/10.4236/jsea.2011.49059>
- Bonin-Font, F., Ortiz, A., & Oliver, G. (2008). Visual navigation for mobile robots: A survey. *Journal of Intelligent and Robotic Systems*, 53(3), 263–296. <https://doi.org/10.1007/s10846-008-9235-4>
- Bragdon, A., Nelson, E., Li, Y., & Hinckley, K. (2011). Experimental analysis of touch-screen gesture designs in mobile environments. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (pp. 403–412). New York, NY: ACM.
- Burnett, G., Crundall, E., Large, D., Lawson, G., & Skrypchuk, L. (2013). A study of unidirectional swipe gestures on in-vehicle touch screens. In *Proceedings of the 5th International Conference on Automotive User Interfaces and Interactive Vehicular Applications* (pp. 22–29). New York, NY: ACM.
- Chandarana, M., Meszaros, E. L., Trujillo, A., & Allen, B. D. (2017). 'Fly like this': Natural language interface for UAV mission planning. *International Conference on Advances in Computer-Human Interactions*. NASA STI.
- Chung, C., & Rantanen, E. Gestural interaction with in-vehicle audio and climate controls. (2010). *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, 54(19), 1406–1410. Sage CA: Los Angeles, CA: SAGE Publications. <https://doi.org/10.1177/154193121005401911>
- Dawes, J. (2008). Do data characteristics change according to the number of scale points used? An experiment using 5-point, 7-point and 10-point scales. *International Journal of Market Research*, 50(1), 61–104. <https://doi.org/10.1177/147078530805000106>
- Demiris, G., Oliver, D. P., Giger, J., Skubic, M., & Rantz, M. (2009). Older adults' privacy considerations for vision based recognition methods of eldercare applications. *Technology and Health Care*, 17(1), 41–48. <https://doi.org/10.3233/THC-2009-0530>
- Deo, N., Rangesh, A., & Trivedi, M. (2016). In-vehicle hand gesture recognition using hidden Markov models. In *2016 IEEE 19th International Conference on Intelligent Transportation Systems (ITSC)* (pp. 2179–2184). Rio de Janeiro, Brazil: IEEE.
- Dovidio, J. F., Brown, C. E., Heltman, K., Ellyson, S. L., & Keating, C. F. (1988). Power displays between women and men in discussions of gender-linked tasks: A multichannel study. *Journal of Personality and Social Psychology*, 55(4), 580. <https://doi.org/10.1037/0022-3514.55.4.580>
- Feng, J., & Donmez, B. (2013). *Designing feedback to induce safer driving behaviors: A literature review and a model of driver-feedback interaction*. Technical report submitted to Toyota Collaborative Safety Research Center (CSRC). Human Factors and Applied Statistics Laboratory, University of Toronto. Retrieved from https://hfast.mie.utoronto.ca/wp-content/uploads/Publications/CSRC_UofT_Report_Literature_review_and_driver_feedback_model.pdf
- Fong, T. W., Conti, F., Grange, S., & Baur, C. (2001). Novel interfaces for remote driving: Gesture, haptic, and PDA. In *Mobile Robots XV and Telemanipulator and Telepresence Technologies VII* (Vol. 4195, pp. 300–311). International Society for Optics and Photonics. <https://doi.org/10.1117/12.417314>
- Forster, Y., Hergeth, S., Naujoks, F., & Krems, J. F. (2018). How usability can save the day-methodological considerations for making automated driving a success story. In *Proceedings of the 10th International Conference on Automotive User Interfaces and Interactive Vehicular Applications* (pp. 278–290). New York, NY: ACM.
- Goldin-Meadow, S. (1999). The role of gesture in communication and thinking. *Trends in Cognitive Sciences*, 3(11), 419–429. [https://doi.org/10.1016/S1364-6613\(99\)01397-2](https://doi.org/10.1016/S1364-6613(99)01397-2)
- González, I. E., Wobbrock, J. O., Chau, D. H., Faulring, A., & Myers, B. A. (2007). Eyes on the road, hands on the wheel: Thumb-based interaction techniques for input on steering wheels. In *Proceedings of Graphics Interface 2007* (pp. 95–102). New York, NY: ACM.
- Harré, N., Field, J., & Kirkwood, B. (1996). Gender differences and areas of common concern in the driving behaviors and attitudes of adolescents. *Journal of Safety Research*, 27(3), 163–173. [https://doi.org/10.1016/0022-4375\(96\)00013-8](https://doi.org/10.1016/0022-4375(96)00013-8)
- Hegarty, M., Montello, D. R., Richardson, A. E., Ishikawa, T., & Lovelace, K. (2006). Spatial abilities at different scales: Individual differences in aptitude-test performance and spatial-layout learning. *Intelligence*, 34(2), 151–176. <https://doi.org/10.1016/j.intell.2005.09.005>
- Holler, J., & Stevens, R. (2007). The effect of common ground on how speakers use gesture and speech to represent size information. *Journal of Language and Social Psychology*, 26(1), 4–27. <https://doi.org/10.1177/0261927X06296428>
- Holler, J., & Wilkin, K. (2009). Communicating common ground: How mutually shared knowledge influences speech and gesture in a narrative task. *Language and Cognitive Processes*, 24(2), 267–289. <https://doi.org/10.1080/016909060802095545>
- Hou, J., Liu, Y., Zheng, T. F., Olsen, J., & Tian, J. (2010). Multi-layered features with SVM for Chinese accent identification. In *2010 International Conference on Audio, Language and Image Processing* (pp. 25–30). Shanghai, China: IEEE.
- Jäger, M. G., Skov, M. B., & Thomassen, N. G. (2008). You can touch, but you can't look: Interacting with in-vehicle systems. In *Proceedings*

- of the SIGCHI Conference on Human Factors in Computing Systems (pp. 1139–1148). New York, NY: ACM.
- Jahani, H., Alyamani, H. J., Kavakli, M., Dey, A., & Billingham, M. (2017). User evaluation of hand gestures for designing an intelligent in-vehicle interface. In *International Conference on Design Science Research in Information System and Technology* (pp. 104–121). Springer, Cham.
- Jaschinski, L., Denjean, S., Petiot, J. F., Mars, F., & Roussarie, V. (2016, October). Impact of interface sonification with touchless gesture command in a car. Human Factors and Ergonomics Society Europe Chapter 2016 Annual Conference, Prague, Czech Republic, pp. 35–46.
- Kendon, A. (2004). *Gesture: Visible action as utterance*. Cambridge University Press.
- Kiefer, P., Giannopoulos, I., Anagnostopoulos, V. A., Schöning, J., & Raubal, M. (2017). Controllability matters: The user experience of adaptive maps. *GeoInformatica*, 21(3), 619–641. <https://doi.org/10.1007/s10707-016-0282-x>
- Kokubo, H., Amano, A., & Hataoka, N. (2002). Robust speech recognition for car environment noise. *Electronics and Communications in Japan (Part III: Fundamental Electronic Science)*, 85(11), 65–73. <https://doi.org/10.1002/ecjc.10055>
- Lee, S. H., Yoon, S. O., & Shin, J. H. (2015). On-wheel finger gesture control for in-vehicle systems on central consoles. In *Adjunct Proceedings of the 7th International Conference on Automotive User Interfaces and Interactive Vehicular Applications* (pp. 94–99). New York, NY: ACM.
- Li, G., Zhang, R., Ritchie, M., & Griffiths, H. (2017). Sparsity-based dynamic hand gesture recognition using micro-Doppler signatures. In *2017 IEEE Radar Conference (RadarConf)* (pp. 0928–0931). Seattle, WA: IEEE. doi:10.1109/RADAR.2017.7944336.
- Liben, L. S., Kastens, K. A., & Stevenson, L. M. (2002). Real-world knowledge through real-world maps: A developmental guide for navigating the educational terrain. *Developmental Review*, 22(2), 267–322. <https://doi.org/10.1006/drev.2002.0545>
- Ma, N., Green, P., Barker, J., & Coy, A. (2007). Exploiting correlogram structure for robust speech recognition with multiple speech sources. *Speech Communication*, 49(12), 874–891. <https://doi.org/10.1016/j.specom.2007.05.003>
- MacEachren, A. M. (2015). Distributed cognition: A conceptual framework for understanding map-based reasoning. In *Proceeding of the 27th International Cartographic Conference*. Rio de Janeiro, Brazil: International Cartographic Association.
- MacKenzie, I. S. (1992). Fitts' law as a research and design tool in human-computer interaction. *Human-Computer Interaction*, 7(1), 91–139. doi:https://doi.org/10.1207/s15327051hci0701_3
- Mashood, A., Noura, H., Jawhar, I., & Mohamed, N. (2015). A gesture based kinect for quadrotor control. In *2015 International Conference on Information and Communication Technology Research (ICTRC)* (pp. 298–301). Abu Dhabi: IEEE. doi:10.1109/ICTRC.2015.7156481
- McNeill, D. (1992). *Hand and mind: What gestures reveal about thought*. University of Chicago Press.
- Millen, D. R. (2000). Rapid ethnography: Time deepening strategies for HCI field research. In *Proceedings of the 3rd Conference on Designing Interactive Systems: Processes, Practices, Methods, and Techniques* (pp. 280–286). New York, NY: ACM.
- Norman, D. (2013). *The design of everyday things: Revised and expanded edition*. Basic Books.
- Ohn-Bar, E., & Trivedi, M. M. (2014). Hand gesture recognition in real time for automotive interfaces: A multimodal vision-based approach and evaluations. *IEEE Transactions on Intelligent Transportation Systems*, 15(6), 2368–2377. <https://doi.org/10.1109/TITS.2014.2337331>
- Pfleging, B., Schneegass, S., & Schmidt, A. (2012). Multimodal interaction in the car: Combining speech and gestures on the steering wheel. In *Proceedings of the 4th International Conference on Automotive User Interfaces and Interactive Vehicular Applications* (pp. 155–162). New York, NY: ACM.
- Pickering, C. A., Burnham, K. J., & Richardson, M. J. (2007). A review of automotive human machine interface technologies and techniques to reduce driver distraction. In *2nd Institution of Engineering and Technology International Conference on System Safety* (pp. 223–228). London, UK. doi: 10.1049/cp:20070468
- Plouffe, G., Cretu, A. M., & Payeur, P. (2015). Natural human-computer interaction using static and dynamic hand gestures. In *2015 IEEE International Symposium on Haptic, Audio and Visual Environments and Games (HAVE)* (pp. 1–6). Ottawa, Canada: IEEE. doi:10.1109/HAVE.2015.7359473
- Reich, D., Weber, M., Terken, J., Riener, A., Schroeter, R., & Osswald, S. (2013). Comparison of different touchless gesture interactions in the car cockpit. In *Adjunct Proceedings of the 5th International Conference on Automotive User Interfaces and Interactive Vehicular Applications* (pp. 119–120). Eindhoven, The Netherlands: ACM.
- Reifinger, S., Wallhoff, F., Ablassmeier, M., Poitschke, T., & Rigoll, G. (2007). Static and dynamic hand-gesture recognition for augmented reality applications. In *International Conference on Human-Computer Interaction* (pp. 728–737). Springer, Berlin, Heidelberg.
- Riener, A., Ferscha, A., Bachmair, F., Hagmüller, P., Lemme, A., Muttenthaler, D., ... Weger, F. (2013). Standardization of the in-car gesture interaction space. In *Proceedings of the 5th International Conference on Automotive User Interfaces and Interactive Vehicular Applications* (pp. 14–21). New York, NY: ACM.
- Rümelin, S., Marouane, C., & Butz, A. (2013). Free-hand pointing for identification and interaction with distant objects. In *Proceedings of the 5th International Conference on Automotive User Interfaces and Interactive Vehicular Applications* (pp. 40–47). New York, NY: ACM.
- Scoditti, A. (2011). A novel taxonomy for testural interaction techniques: Considerations for automotive environments. In *3rd International Workshop on Multimodal Interfaces for Automotive Applications (MIAA)*. Palo Alto, CA.
- Singha, J., Roy, A., & Laskar, R. H. (2018). Dynamic hand gesture recognition using vision-based approach for human-computer interaction. *Neural Computing & Applications*, 29(4), 1129–1141. <https://doi.org/10.1007/s00521-016-2525-z>
- Stern, H. I., Wachs, J. P., & Edan, Y. (2008). Designing hand gesture vocabularies for natural interaction by combining psycho-physiological and recognition factors. *International Journal of Semantic Computing*, 2(1), 137–160. <https://doi.org/10.1142/S1793351X08000385>
- Sun, X., & Li, T. (2014). *Survey of studies supporting the use of in-air gesture in car HMI. UXPA (China)*. Wuxi, Jiangsu, China.
- Terada, T., Miyamae, M., Kishino, Y., Tanaka, K., Nishio, S., Nakagawa, T., & Yamaguchi, Y. (2006). Design of a car navigation system that predicts user destination. In *7th International Conference on Mobile Data Management (MDM'06)* (pp. 145). Nara, Japan: IEEE.
- Wigdor, D., & Wixon, D. (2011). *Brave NUI world: Designing natural user interfaces for touch and gesture*. Elsevier.
- Wolf, K., Naumann, A., Rohs, M., & Müller, J. (2011). A taxonomy of microinteractions: Defining microgestures based on ergonomic and scenario-dependent requirements. In *IFIP Conference on Human-Computer Interaction* (pp. 559–575). Springer, Berlin, Heidelberg.
- Wu, H., Zhang, S., Liu, J., Qiu, J., & Zhang, X. (2019). The gesture disagreement problem in free-hand gesture interaction. *International Journal of Human-Computer Interaction*, 35(12), 1102–1114. <https://doi.org/10.1080/10447318.2018.1510607>
- Wu, S., Gable, T., May, K., Choi, Y. M., & Walker, B. N. (2016). Comparison of surface gestures and air gestures for in-vehicle menu navigation. *Archives of Design Research*, 29(4), 65–80. <https://doi.org/10.15187/adr.2016.11.29.4.65>
- Yan, Q., & Vaseghi, S. (2002). A comparative analysis of UK and US English accents in recognition and synthesis. In *2002 IEEE International Conference on Acoustics, Speech, and Signal Processing* (Vol. 1, pp. 1–413). Orlando, FL: IEEE.
- Yin, Y., & Davis, R. (2014). Real-time continuous gesture recognition for natural human-computer interaction. In *2014 IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC)* (pp. 113–120). Melbourne, VIC, Australia: IEEE.
- Zeng, J., Sun, Y., & Wang, F. (2012, November). A natural hand gesture system for intelligent human-computer interaction and medical assistance. In *2012 Third Global Congress on Intelligent Systems* (pp. 382–385). Wuhan, China: IEEE.
- Zhuang, H., Yang, M., Cui, Z., & Zheng, Q. (2017). A method for static hand gesture recognition based on non-negative matrix factorization and

compressive sensing. *IAENG International Journal of Computer Science*, 44 (1), 52–59. http://www.iaeng.org/IJCS/issues_v44/issue_1/index.html and the article's direct pdf link is: http://www.iaeng.org/IJCS/issues_v44/issue_1/IJCS_44_1_07.pdf

About the Authors

Xiaosong Qian is a PhD design researcher at Beijing Institute of Technology (BIT) and a Visiting Research Fellow at Stanford University's Center for Design Research. His research focuses on user interaction, user interfaces and user-centered design. He has published several papers in international journals and conference proceedings.

Wendy Ju is an Associate Professor of Information Science at Cornell Tech. She works on designing interactions with automation, with a focus on human–robot interaction and autonomous vehicle interaction. She holds a PhD in Mechanical Engineering from Stanford, and a MS in Media Arts & Sciences from MIT (2008).

David Sirkin is Executive Director of Interaction Design Research at Stanford's Center for Design Research. His research focuses on the design of physical interactions between humans and robots, and autonomous vehicles and their interfaces. He holds a PhD in Mechanical Engineering (Design) from Stanford, and MS degrees in EECS and Management from MIT.